RICE TYPE CLASSIFICATION

A Machine Learning Approach to Grain Morphology

Seminar Presentation • Karthick

THE DATA

Dataset Overview

The dataset contains 18,185 grains of rice, each described by 10 numeric shape and size measurements extracted from grain images.

Two rice classes are compared:

- Class 0 — 8,200 grains
- Class 1 — 9,985 grains

Features include Area, Major/Minor Axis Length, Eccentricity, Convex Area, Equivalent Diameter, Extent, Perimeter, Roundness, and Aspect Ratio — all describing grain shape.

The goal: build a model that predicts rice Class purely from these measurements.

18,185
total rice grains analyzed

12
columns in dataset

2
rice classes

10
shape features

80 / 20
train / test split

Shape Features of a Rice Grain

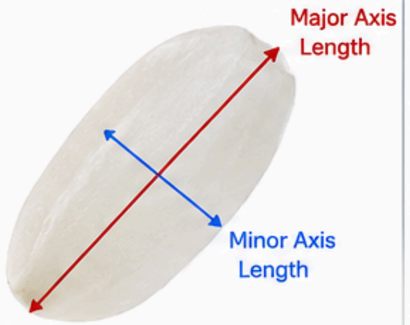
These features are commonly used in image analysis to quantify the shape characteristics of rice grains.

1. Area



Total number of pixels within the boundary of the grain.

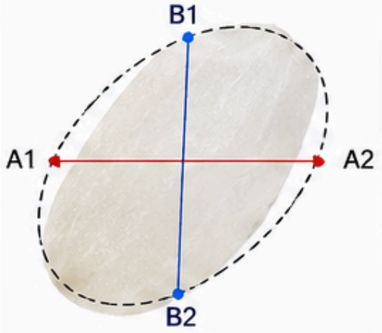
2. Major / Minor Axis Length



Major Axis Length: Length of the longest line passing through the grain.

Minor Axis Length: Length of the longest line perpendicular to the major axis.

3. Eccentricity



Eccentricity (e) = $\frac{c}{a}$

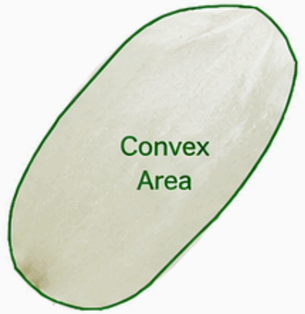
Where,

c = distance from center to a focus (A1 or A2)

a = length of the semi-major axis (A1Æ/2)

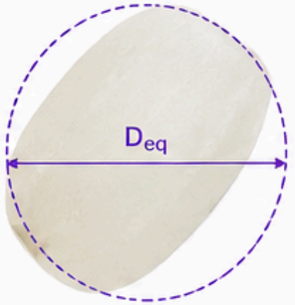
Range: 0 (circle) to 1 (line)

4. Convex Area



Area of the smallest convex polygon that completely encloses the grain.

5. Equivalent Diameter



Diameter of a circle having the same area as the grain.

$$D_{eq} = \sqrt{\frac{4A}{\pi}}$$

6. Extent



Ratio of the grain area to the area of its bounding rectangle.

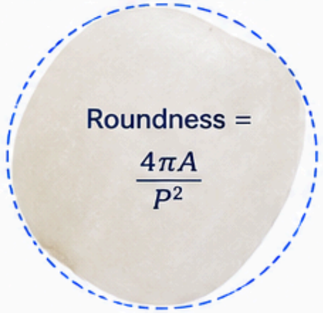
$$\text{Extent} = \frac{\text{Area of grain}}{\text{Area of bounding rectangle}}$$

7. Perimeter



Total length of the boundary of the grain.

8. Roundness



Measures how close the shape is to a perfect circle.

Range: 0 (not round) to 1 (perfectly round)

9. Aspect Ratio



Ratio of Major Axis Length to Minor Axis Length.

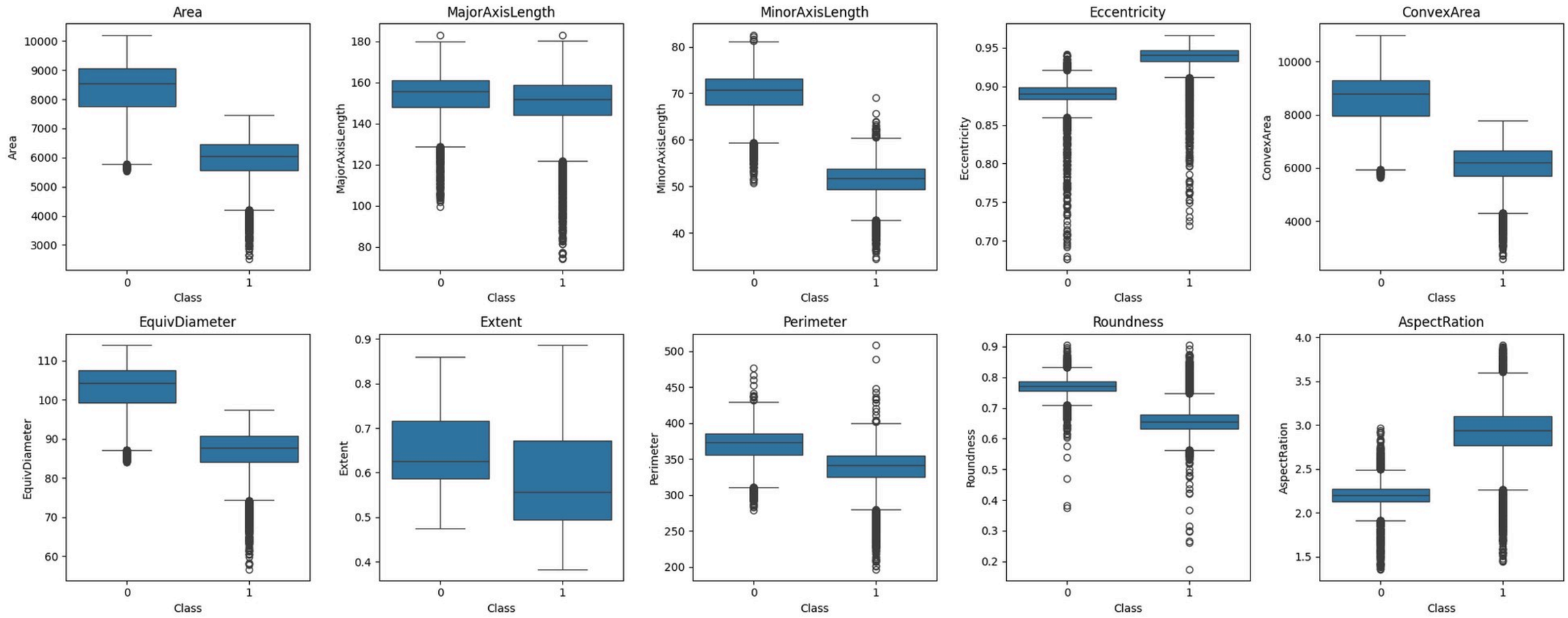
$$\text{Aspect Ratio} = \frac{\text{Major Axis Length}}{\text{Minor Axis Length}}$$

Example Rice Grain



EXPLORATORY ANALYSIS

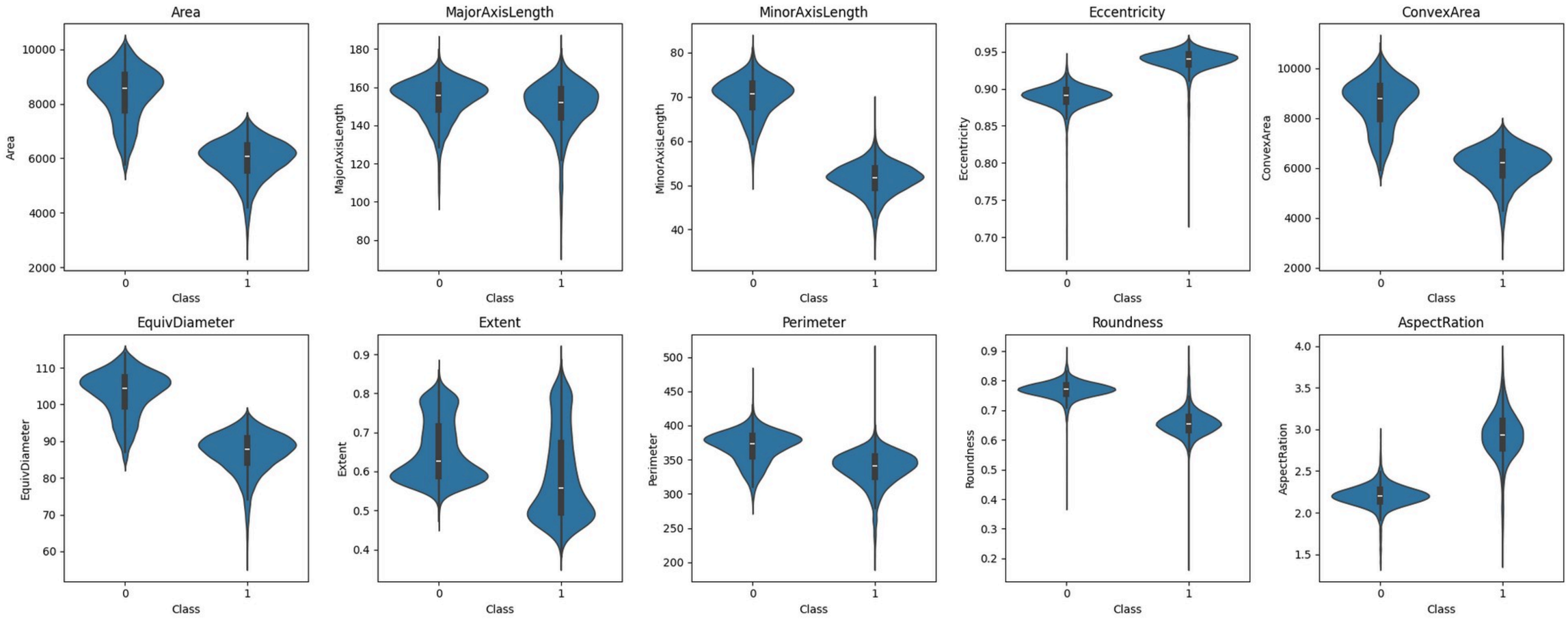
Boxplots by Class



Splitting each feature by Class confirms clear separation: Class 0 grains are consistently larger in Area, MinorAxisLength, ConvexArea and EquivDiameter, while Class 1 grains score higher on AspectRatio and Eccentricity.

EXPLORATORY ANALYSIS

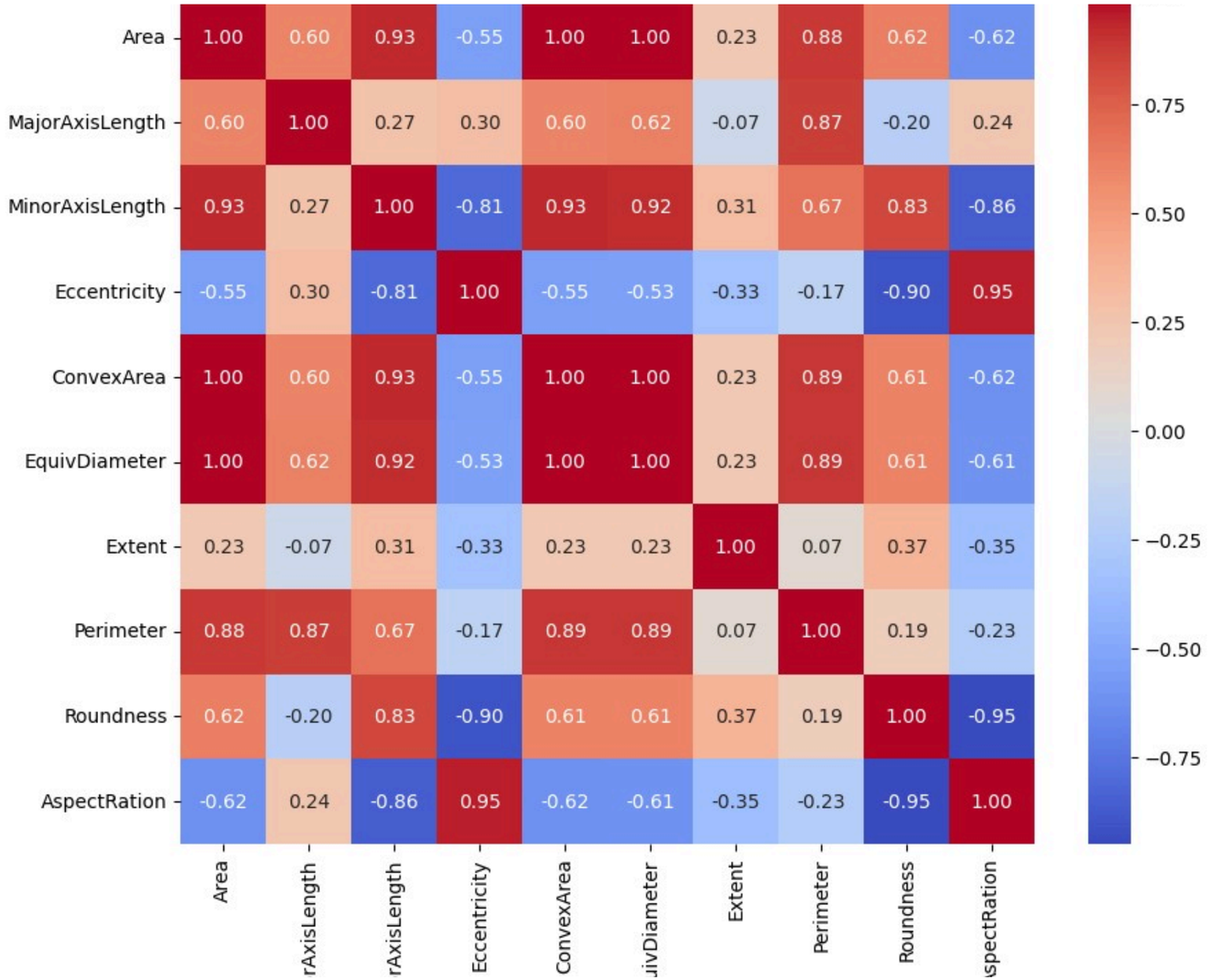
Violin Plots by Class



Violin plots add the shape of each distribution on top of the boxplot summary. Most features show two distinct, minimally-overlapping "bulges" per class — visual proof that these features can separate the classes well.

EXPLORATORY ANALYSIS

Correlation Heatmap



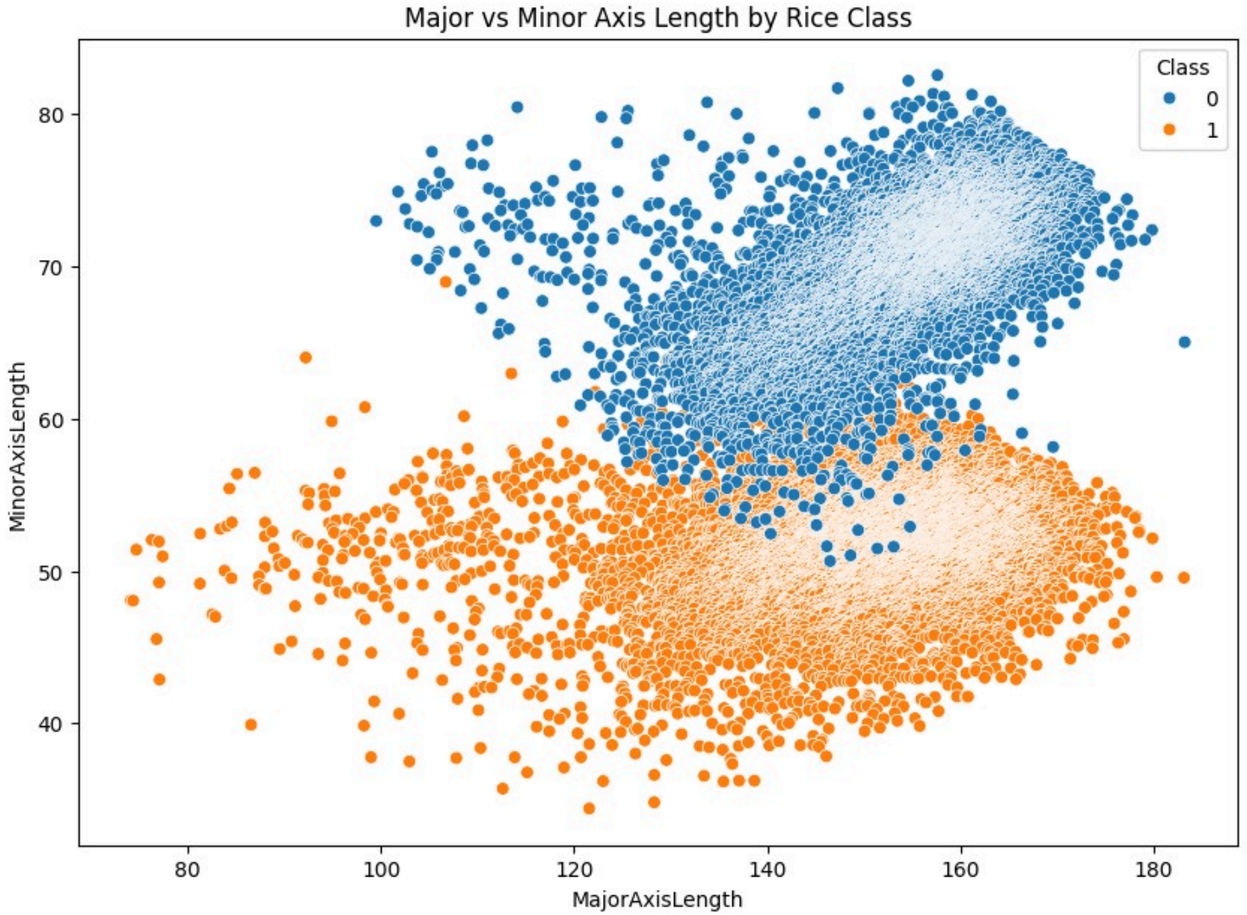
Area, ConvexArea, and EquivDiameter are almost perfectly correlated (0.99–1.00) — they all measure grain size in different units, so they carry overlapping information.

Roundness and AspectRatio are strongly negatively correlated (-0.95): round grains have a low aspect ratio, elongated grains have a high one.

This tells us some features are redundant and could be trimmed without losing much information.

EXPLORATORY ANALYSIS

Major vs Minor Axis Length



Plotting MajorAxisLength against MinorAxisLength, colored by Class, shows two clearly separated clusters with only a thin overlap zone.

Class 0 (blue) grains are shorter and wider. Class 1 (orange) grains are longer and thinner — a higher aspect ratio.

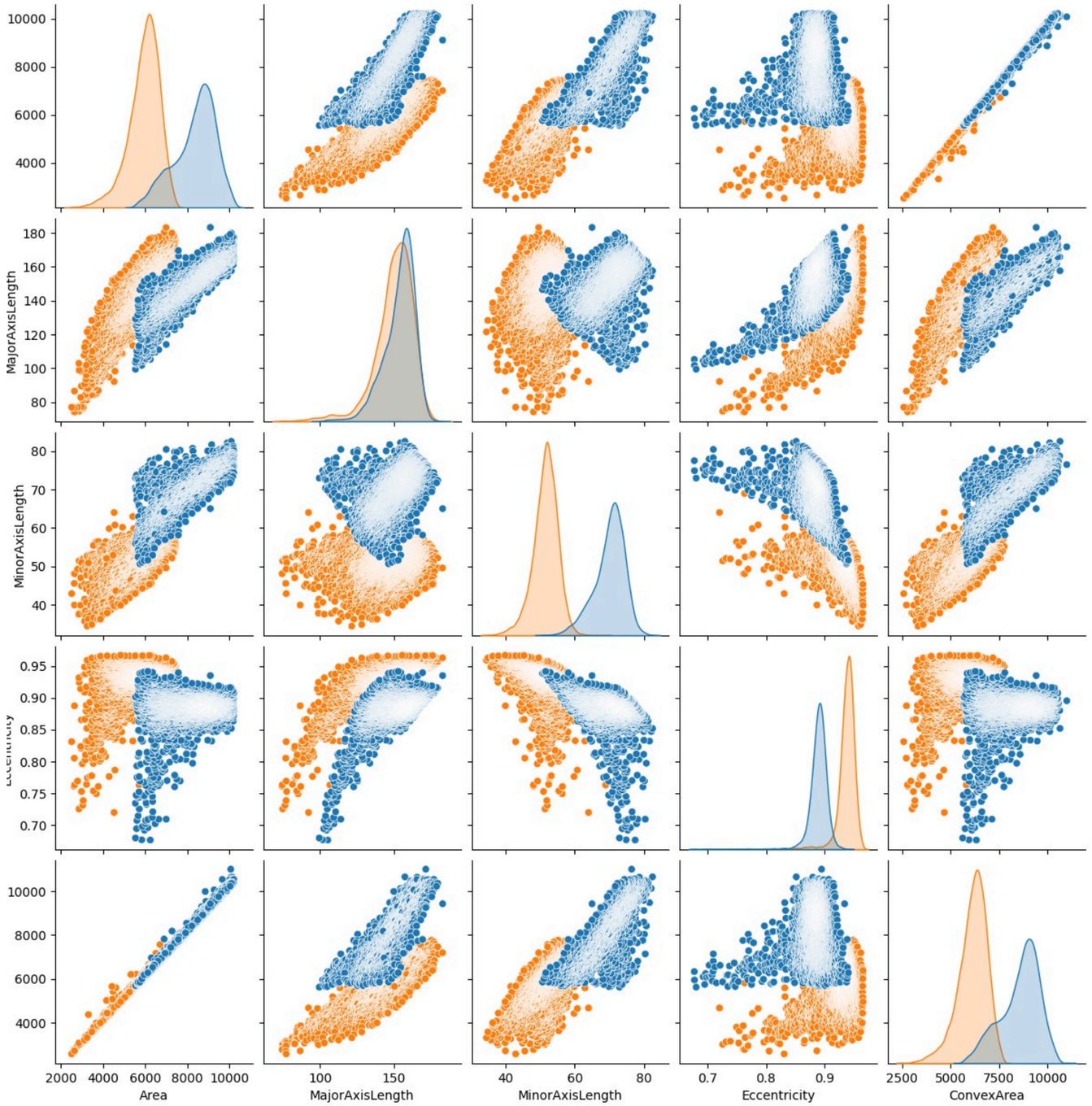
This single pair of features already does most of the separating work.

EXPLORATORY ANALYSIS

Pairplot of Key Features

The pairplot shows every pairwise combination of the first five features, colored by Class, with each feature's own distribution on the diagonal.

Nearly every scatter panel shows two separable clouds of points — reinforcing that this is a genuinely learnable classification problem, not just noise.



DATA SCALING STRATEGY

3. FEATURE SCALING — StandardScaler

The 10 features are on very different scales — this would bias distance-based models like SVM and KNN toward whichever feature has the biggest raw numbers.

BEFORE SCALING

Area, ConvexArea

≈ 3,000 – 10,000

Eccentricity, Roundness, Extent

≈ 0.0 – 1.0

Scaler →

AFTER SCALING

Every feature transformed to:

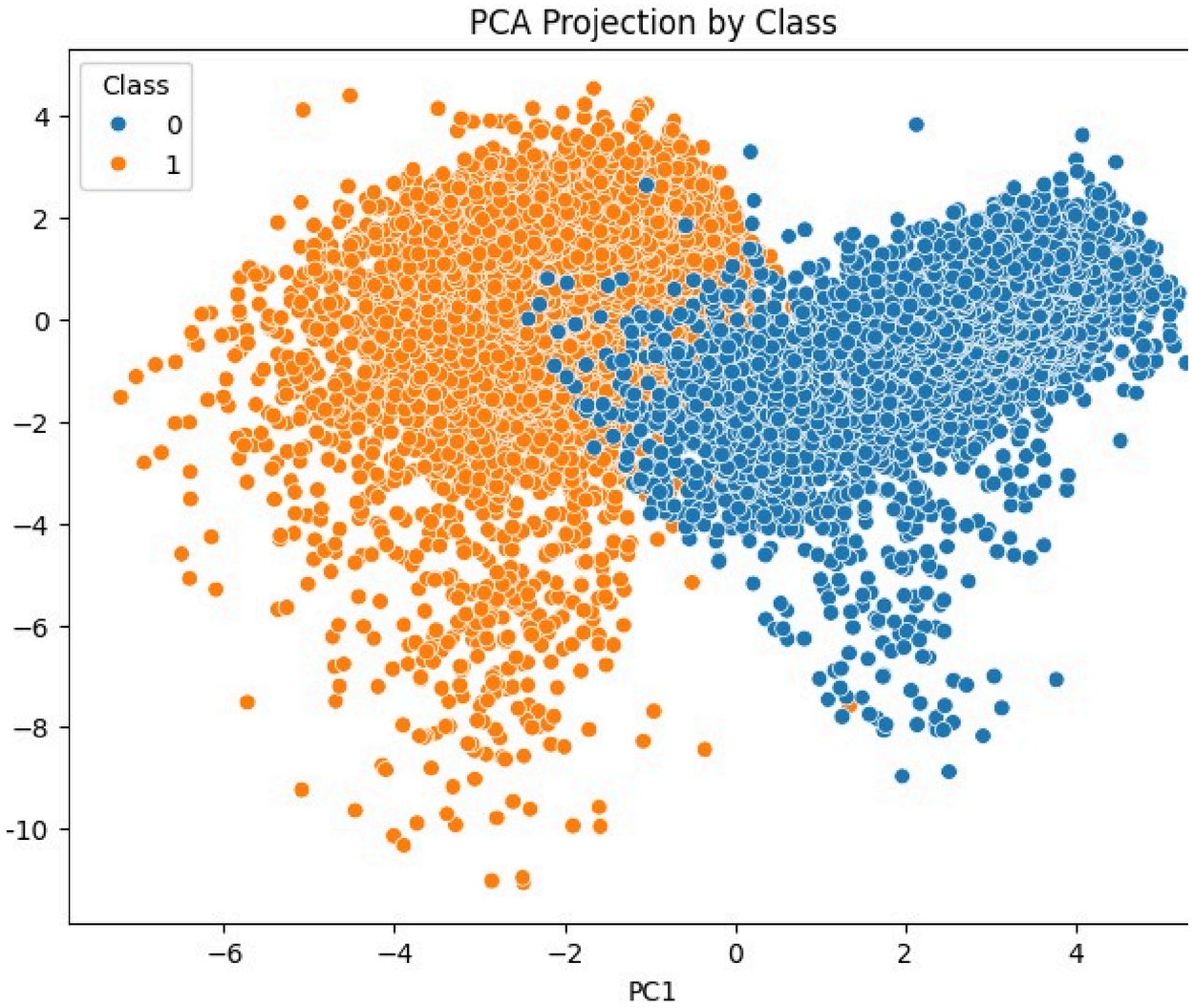
Mean = 0 **Std Dev = 1**

$$\text{scaled value} = (\text{original value} - \text{mean}) / \text{standard deviation}$$

fit_transform() was applied only on the training set; transform() (not re-fit) was then applied to the test set — this prevents information from the test set leaking into training.

DIMENSIONALITY REDUCTION

PCA Projection



Principal Component Analysis compresses all 10 (scaled) features into 2 components while keeping as much information as possible.

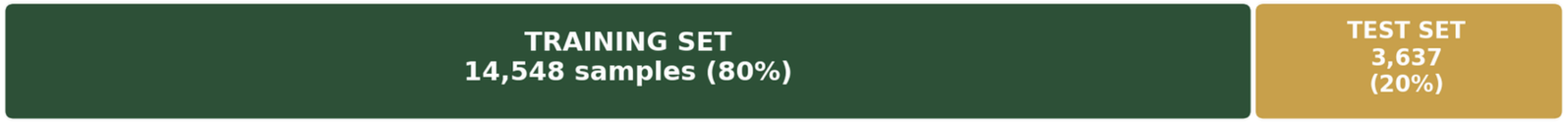
Even in just 2 dimensions, the two rice classes form two mostly-separate clouds with only a small overlapping boundary region.

This confirms the classes are separable using simple, mostly-linear boundaries — which is why even basic models perform extremely well later.

TRAINING & TESTING

1. TRAIN / TEST SPLIT

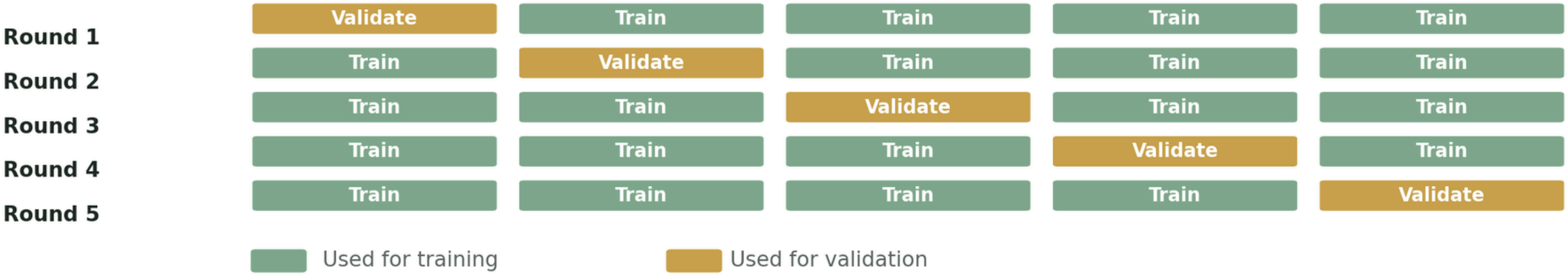
Total dataset: 18,185 rice grains → split 80% / 20%



The model learns patterns only from the training set, then is evaluated on the test set — grains it has never seen — to check if it generalizes correctly.

2. 5-FOLD CROSS-VALIDATION

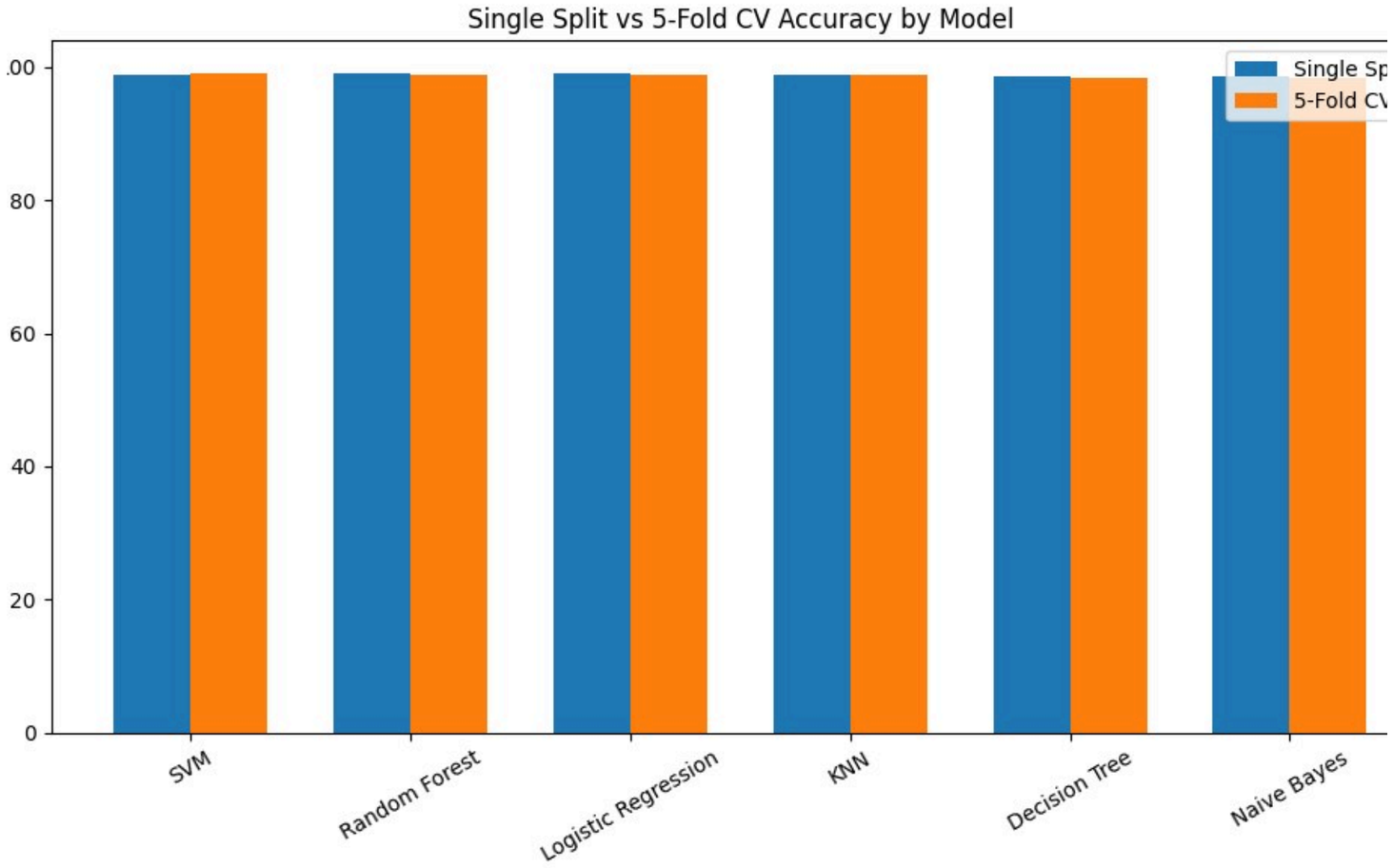
The 14,548 training samples are split into 5 equal folds. The model trains 5 times, using a different fold as validation each round — so every sample gets tested exactly once.



The 5 accuracy scores are then averaged — giving a more reliable measure than a single train/test split, since every sample gets a turn being tested.

MACHINE LEARNING RESULTS

Model Comparison



Model	Single Split	5-Fold CV
SVM	98.90%	**99.04%**
Random Forest	98.98%	98.96%
Logistic Regression	98.98%	98.91%
KNN	98.82%	98.82%
Decision Tree	98.60%	98.36%
Naive Bayes	98.60%	98.32%

Six models were compared using both a single 80/20 split and 5-fold cross-validation. All models score above 98% — the features are highly informative. SVM performs best on cross-validation at 99.04%.

MACHINE LEARNING RESULTS

Confusion Matrix

Total test samples: 3,637

Correctly identified: 3,597

Misclassified: 40

Predicted Class 0

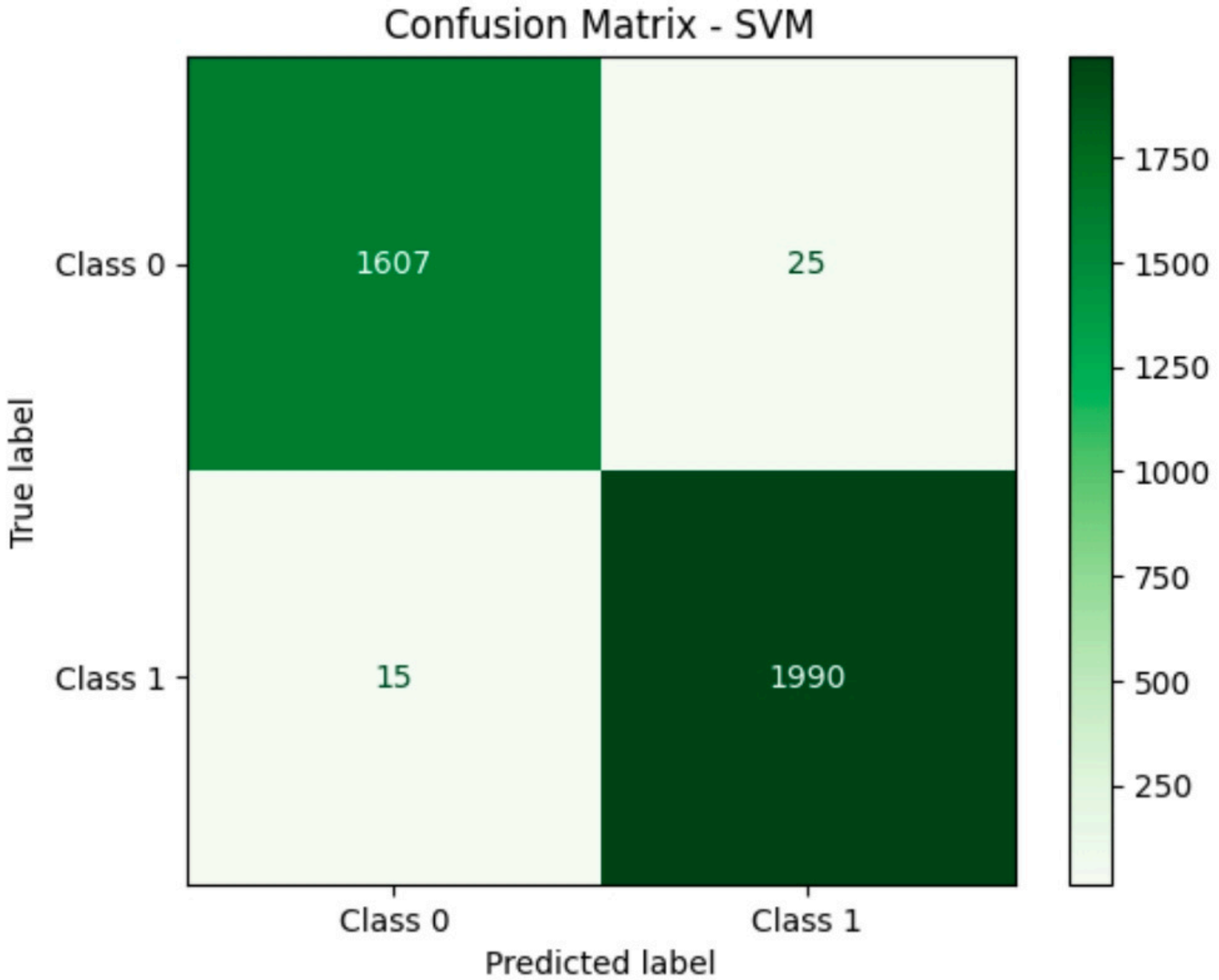
- 1,607 ✓
- 25 ✗

Actual Class 0 (1,632 grains)

Predicted Class 1

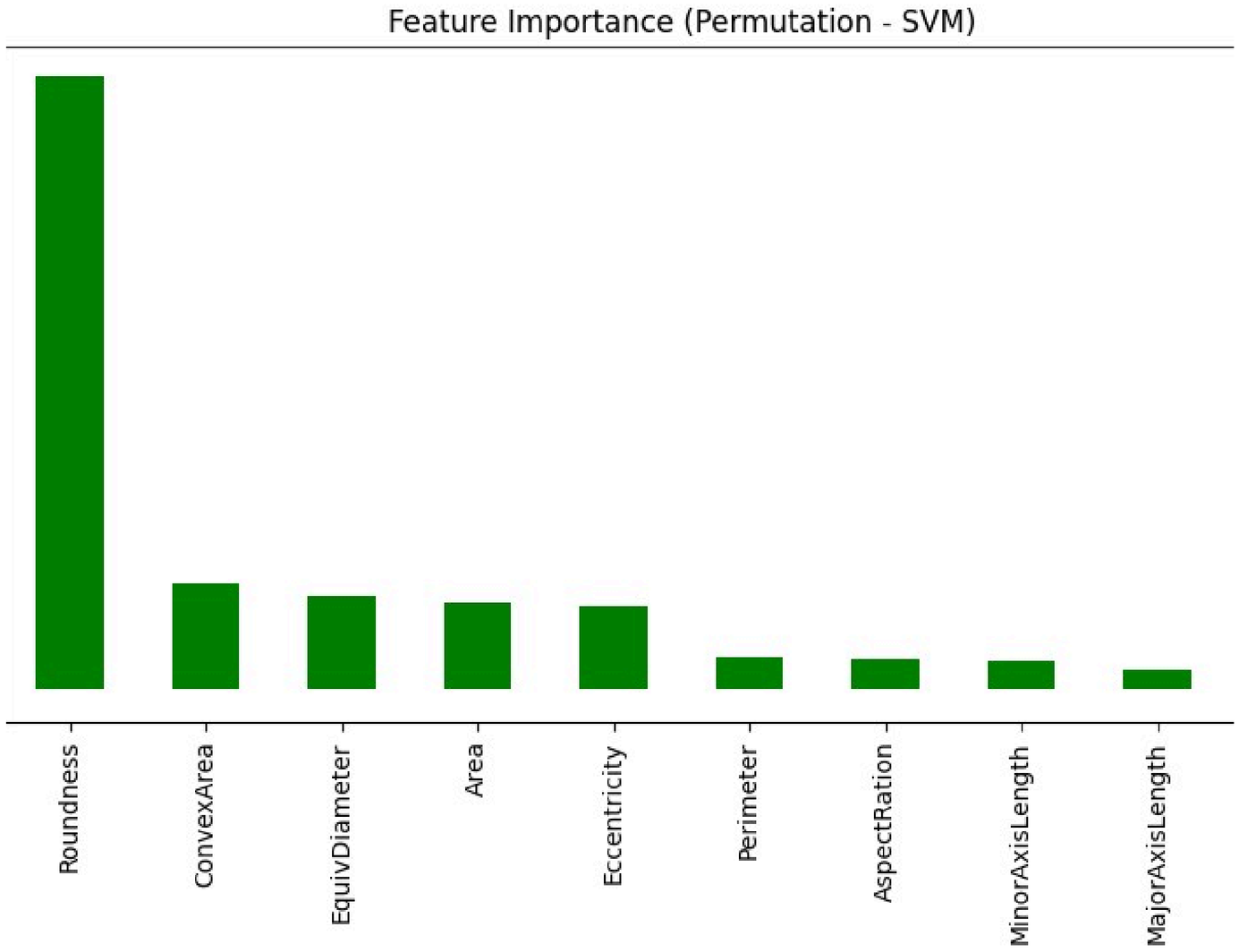
- 1990 ✓
- 15 ✗

Actual Class 1 (2,005 grains)



MACHINE LEARNING RESULTS

Feature Importance



Permutation importance on the SVM model measures how much accuracy drops when a feature's values are randomly shuffled.

Roundness stands out as by far the most important feature — shuffling it alone causes the biggest accuracy drop.

ConvexArea, EquivDiameter, Area, and Eccentricity contribute moderately. Extent contributes almost nothing, meaning it could likely be dropped from the model.

CONCLUSION

Grain Shape Alone Predicts Rice Type

- The two rice classes differ clearly in shape — mainly Roundness, AspectRatio, and grain size.
- Exploratory plots (histograms, boxplots, PCA) all showed separable clusters before any model was built.
- All 6 machine learning models achieved 98%+ accuracy, with SVM performing best (99.04% CV accuracy).
- Roundness was the single most important feature for prediction; Extent added almost nothing.
- Cross-validation confirmed the results were stable, not a lucky single split.